

TriPoDPy: 1D Tri-Population size distributions for Dust evolution in protoplanetary disks

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Software

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Summary

TriPoDPy is a code simulating the dust evolution, including dust growth and dynamics in
protoplanetary disks using the parametric dust model presented in (Pfeil et al., 2024). The
simulation evolves a dust distribution in a one-dimensional grid in the radial direction. It's
written in Python and the core routines are implemented in Fortran90. The code not only
solves for the evolution of the dust but also the gas disk with the canonical α -description
(Shakura & Sunyaev, 1973). In addition to the original model, we added descriptions of tracers
for the dust and gas, which could be used for compositional tracking of additional components.

Statement of Need

Simulating the dust evolution in protoplanetary disks, including growth and transport, is vital to
understanding planet formation and the structure of protoplanetary disks. TriPoDPy describes
the vertically integrated dust size distribution with a truncated power law, which allows the
simulation full access to the dust size distribution, which is essential to accurately model the
dust evolution and additional physical effects like photoevaporation. Additionally, TriPodPy
enables the addition of tracers in gas and dust, which could be used for tracking of chemical
composition, electrical charge, and other parameters. All these features make it a valuable
tool for astronomers to quickly model the dust evolution of protoplanetary disks.

State of the Field

There exist multiple open-source codes that tackle the dust evolution in Protoplanetary disks
by either solving the Smoluchowski Equation, e.g. Dustpy (S. M. Stammer & Birnstiel,
2022) or CuDisc (Robinson et al., 2024) or using a Monte Carlo approach (e.g. Mcdust
(Vaikundaraman et al., 2025)) to simulate the mutual collisions between dust grains. However,
all these simulations are computationally expensive, which calls for parametrised dust evolution
models that can be used, for example, for population studies. Previous parametric models,
e.g. Twopoppy (Birnstiel et al., 2012), were not designed for disks with radial sub-structures
and were not calibrated for different stellar masses. These shortcomings are solved with the
TriPoD Dust model.

Software design

TriPoDPy builds on the Simframe (S. Stammer & Birnstiel, 2022) framework to allow users
to easily modify and use our code by handling file IO, etc. To increase performance, the core

38 routines have been implemented in Fortran90. Since we are solving the same problem as the
39 full Smoluchowski solver Dustpy, our code inherits several shared routines, making it easy for
40 users familiar with Dustpy to adopt the code and convert setups between the two codes.

41 Research impact statement

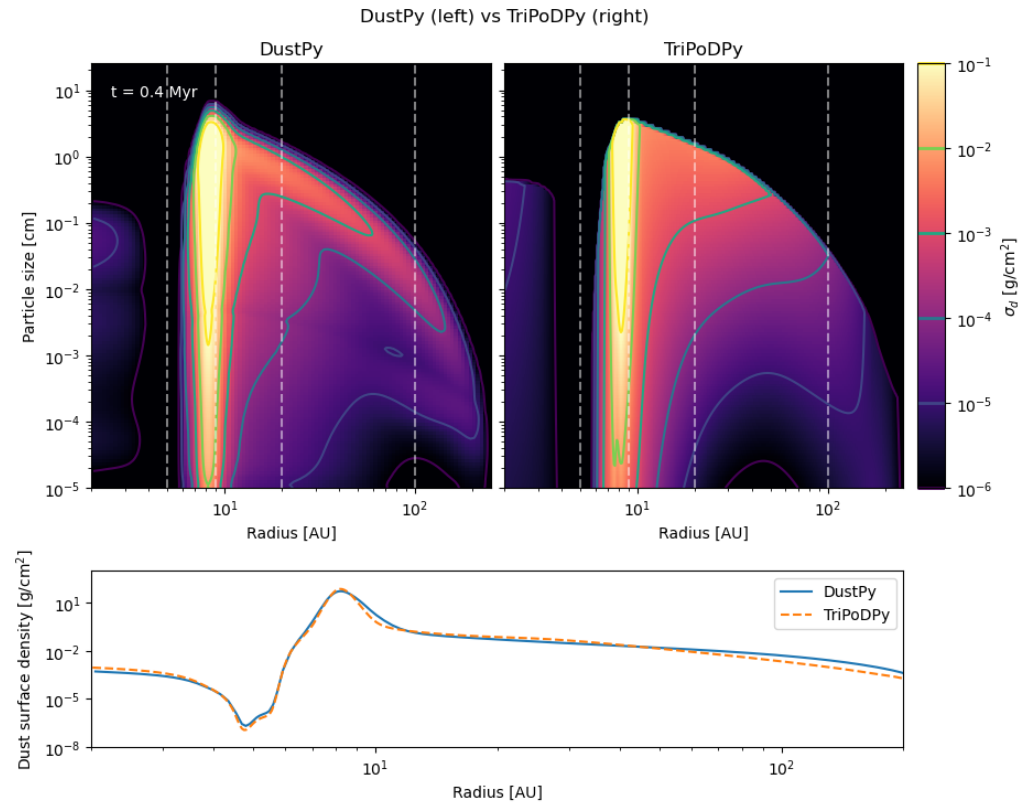
42 Currently, there are multiple ongoing projects that use this code, with the first results already
43 in preprint (Ziampras et al., 2026). Since the code is a straight upgrade over the widely used
44 predecessor Twopoppy (Birnstiel et al., 2012) (with over 700 citations) for most cases, we
45 expect many projects to make use of our code.

46 Comparison Simulation

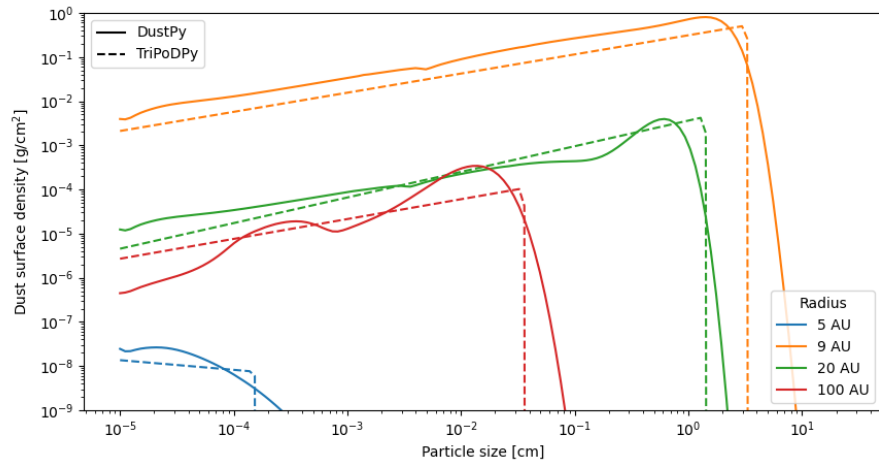
47 We compare a Simulation with our code with one performed with the full coagulation code
48 Dustpy, illustrating how well our code performs. The parameters used for the comparison
49 simulations can be found in the Table below:

Parameter	Value
gas surface density at 1 AU	722 g/cm ²
temperature at 1 AU	209 K
turbulence strength (α)	10 ⁻³
fragmentation velocity (v_{frag})	10 m/s
gas surface density power law p	0.85
temperature power law q	0.5
gap position	5.2 AU
$M_{\text{planet}}/M_{\star}$	10 ⁻³

50 We compare the particle size distribution from both simulations at 400'000 years, which can
51 be seen in the figures below. The first figure shows the dust surface density as a function of
52 size and radius throughout the disk (top) and the total dust surface density as a function of
53 radius (bottom). The second plot shows 1-D slices at different radii as indicated by the white
54 dashed lines.



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The TriPoDPy simulation runs a factor of 50 to 100 faster than the compared DustPy model. As we can see, the maximal sizes and dust size distributions match quite well with the full coagulation code. Since the size distribution is always assumed to be a power law, capturing multimodal distributions is not possible, as can be seen around 100 AU in the test simulation. This also affects the dust distribution on the inside of the gap, as the dust size distribution in gaps deviates from the expected power law as well. For an in-depth discussion, see (Pfeil et al., 2024).

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71 held responsible for them.

72 AI usage disclosure

73 AI (GitHub Copilot) was used to generate implementation ideas for the code, however all
74 prompts were subsequently checked and corrected manually.

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